A genome-wide CRISPRa targetability atlas across murine microglial states integrating PAM availability with chromatin accessibility

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Abstract

Background: CRISPR-mediated transcriptional activation (CRISPRa) delivered by adeno-associated virus (AAV) is an emerging therapeutic modality for neurological diseases. Single-AAV delivery requires compact Cas proteins—such as Un1Cas12f1 (HEAL), Un1Cas12f1 (Smin-iCRa), and CjCas9 (MiniCAFE)—whose protospacer-adjacent motif (PAM) sequences are less frequent than SpCas9's NGG. It is widely assumed that this PAM restriction substantially limits the fraction of therapeutically relevant genes that can be targeted by single-AAV CRISPRa systems. However, no systematic evaluation of CRISPRa targetability has been performed that integrates PAM availability with cell-type-specific chromatin accessibility.

Results: We constructed a genome-wide CRISPRa targetability atlas for murine microglia encompassing 21,599 protein-coding gene promoters, six Cas orthologs, and six microglial functional states from three independent laboratories (homeostatic, naïve, acute inflammatory, tolerized, post-surgical sham, and post-ischemic stroke). Within the promoter window and filtering scheme used here, PAM availability alone was high: even the most restrictive PAM (TTTR, Un1Cas12f1) yielded $\geq$1 suitable protospacer in 89.1% of 55 curated therapeutic genes after strict IUPAC-aware scanning on both strands. Integrating ATAC-seq-derived chromatin accessibility substantially reduced predicted promoter-level targetability across the six Cas orthologs to 32.7–40.0% in homeostatic microglia, 45.5–58.2% in inflammatory states, and 60.0–74.5% in surgical/stroke-context states. Within each state, the spread across Cas orthologs was 7.3–10.9 percentage points, smaller than the 32.7 percentage-point range observed across microglial states for HEAL. The *Tfeb* promoter was classified as accessible only in post-surgical sham microglia under the atlas midpoint criterion, whereas its paralog *Tfe3* was classified as accessible across all six states. Thirteen therapeutic genes showed surgical/stroke-context accessibility patterns that should not be interpreted as injury-driven without validation.

Conclusions: This study provides a predictive in silico resource for promoter-targeted CRISPRa design in murine microglia. The most direct contribution is a transparent, state-aware atlas showing that chromatin accessibility can substantially narrow the set of promoter-level CRISPRa candidates after PAM scanning. The predictions should be interpreted as hypotheses for experimental validation, not as demonstrated CRISPRa activity or therapeutic efficacy.

Keywords: CRISPRa, microglia, ATAC-seq, chromatin accessibility, AAV, Cas12f, PAM, targetability atlas

1 Background

CRISPR-mediated transcriptional activation (CRISPRa) enables upregulation of endogenous genes without permanent genome modification, making it an attractive modality for in vivo therapeutic gene regulation [1]. When delivered by adeno-associated virus (AAV), CRISPRa systems must contend with the $\sim$4.7 kb packaging capacity between the inverted terminal repeats (ITRs), which constrains the size of the nuclease-dead Cas protein, the transcriptional activation domain, the promoter, the single-guide RNA (sgRNA) cassette, and regulatory elements.

The most widely used Cas protein, SpCas9, exceeds 4.1 kb in coding sequence alone, precluding single-AAV packaging of dSpCas9-based CRISPRa constructs with potent activation domains such as VPR or miniVPR [2]. This has driven the development of compact Cas alternatives compatible with single-AAV delivery. Among these, Un1Cas12f1 ($\sim$1.6 kb; HEAL system) [3], Un1Cas12f1 ($\sim$1.6 kb; SminiCRa) [4], and CjCas9 ($\sim$2.9 kb; MiniCAFE) [5] have demonstrated CRISPRa activity in mammalian cells. HEAL and SminiCRa are both based on Un1Cas12f1 but use different engineered sgRNA scaffolds and activation domains. These compact Cas proteins recognize PAM sequences that are less frequent than SpCas9's NGG: TTTR (R = A or G; i.e., TTTA or TTTG) for Un1Cas12f1-based systems, and NNNVRYM for CjCas9. It is widely assumed—though not systematically quantified—that these PAM restrictions limit the fraction of therapeutically relevant genes that can be targeted.

A second, largely overlooked constraint is chromatin accessibility. CRISPRa requires that the sgRNA-Cas complex physically access the target promoter, which in turn requires that the DNA be in an open chromatin state at the time of transduction. Chromatin accessibility is not a fixed property of a locus but varies dramatically across cell types and functional states [6, 7]. In microglia—the resident macrophages of the central nervous system and an emerging target for gene therapy in neurodegeneration, spinal cord injury (SCI), and neuroinflammation—chromatin landscapes shift substantially between homeostatic, developmental, inflammatory, and disease-associated states [8–13].

Prior work has established that chromatin accessibility influences CRISPRa efficacy and should be incorporated into sgRNA design. Horlbeck et al. integrated DNase-seq and MNase-seq data into genome-scale CRISPRi/a library design [14], subsequent screens have refined CRISPRa library design [15, 16], and computational tools such as crisprVerse [17] support chromatin-aware guide selection. However, these resources provide general-purpose design algorithms rather than cell-type-specific targetability maps. Despite the growing interest in CRISPRa-based approaches targeting microglia [18], no resource has applied these principles to systematically evaluate predicted promoter-level targetability across microglial functional states, including disease-relevant contexts such as CNS injury. Here, we extend existing chromatin-aware CRISPRa design methodology to generate a microglia-focused targetability atlas, integrating PAM availability for six Cas orthologs with ATAC-seq-derived chromatin accessibility across six microglial states from three independent laboratories. The goal is not to introduce a new general principle for CRISPRa design, but to provide a transparent, state-aware resource for prioritizing hypotheses before experimental validation.

2 Results

2.1 PAM availability alone is high across therapeutic gene promoters

To evaluate the contribution of PAM restriction to CRISPRa targetability, we scanned promoter regions of all 21,599 protein-coding genes in the mouse genome (GENCODE vM33, GRCm39) for

PAM sequences recognized by six Cas orthologs spanning a range of PAM specificities and AAV packaging compatibilities (Table 1). For each gene, we defined the CRISPRa-optimal promoter window as −400 to −50 bp relative to the annotated transcription start site (TSS), corresponding to the region where CRISPRa systems show maximal transcriptional activation [1]. We scanned both DNA strands for PAM sequences and applied basic quality filters to the resulting protospacers: GC content 30–70%, absence of homopolymer runs >4 nt, and absence of restriction enzyme sites used in standard cloning (BsmBI, BsaI).

Across the full genome, PAM availability was high for all Cas orthologs. SpCas9 (NGG) identified $\geq$1 passing protospacer in 99.98% of gene promoters (21,594/21,599), and the most restrictive PAM—TTTR of Un1Cas12f1 (shared by HEAL and SminiCRa)—covered 93.3% of promoters (20,160/21,599; Figure 2A). When focused on a curated panel of 55 therapeutic genes, coverage remained high: HEAL/SminiCRa 89.1% (49/55), SaCas9 96.4% (53/55), and SpCas9, CjCas9, and Nme2Cas9 100% (55/55) (Figure 2B). Six therapeutic genes (*Cx3cr1*, *Dnmt3a*, *Hdac2*, *Pparg*, *Tet2*, and *Tmem119*) lacked a passing TTTR protospacer in the CRISPRa-optimal window.

Notably, CjCas9's PAM (NNNVRYM), which at 7 nt is the longest in our panel, achieved 100% coverage. This result arises because three of the seven PAM positions are fully degenerate (N) and the remaining constrained positions are themselves degenerate (V, R, Y, and M), yielding an expected frequency of approximately 1 per 11 bp per strand and therefore many candidate sites in a 350 bp window.

For *Tfeb* (transcription factor EB, master regulator of autophagy-lysosomal biogenesis [19, 20]), we identified 5 passing TTTR PAM sites for HEAL/SminiCRa, 23 for SpCas9, and 6 for SaCas9 within the CRISPRa-optimal window. Although fewer than SpCas9, these TTTR sites provide multiple candidate positions for downstream sgRNA evaluation.

These results indicate that, within the 350 bp promoter window evaluated here, PAM availability alone provides broad coverage of the curated therapeutic gene panel, including for compact single-AAV-compatible Cas systems.

2.2 ATAC-seq filtering reduces predicted targetability

To evaluate the functional constraint of chromatin state on CRISPRa targetability, we uniformly reprocessed publicly available ATAC-seq datasets from three independent studies covering six microglial functional states: homeostatic adult microglia (ex vivo purified; Gosselin et al., GSE89960 [8]), naïve (PBS control; Zhang X et al., GSE175578 [13]), acute LPS-stimulated (PBS then LPS; GSE175578), LPS-tolerized (LPS then LPS; GSE175578), post-surgical sham (FACS-sorted microglia from sham-operated mice; Zhang et al., GSE220041 [21]), and post-ischemic stroke (transient middle cerebral artery occlusion, tMCAO; GSE220041). All datasets were aligned to GRCm39 with Bowtie2 [22], PCR duplicates removed with Picard MarkDuplicates, filtered (MAPQ≥30, chrM removed, ENCODE blacklist regions excluded [23]), and peaks called with Genrich (ATAC-seq mode) [24].

We then intersected PAM positions from Phase 1 with ATAC-seq peaks from each microglial state. A gene was classified as *predicted targetable* for a given Cas ortholog in a given state if: (1) its promoter region overlapped an ATAC-seq peak under the atlas accessibility criterion, and (2) at least one PAM site with a passing protospacer fell within an accessible region.

The integration of chromatin accessibility data substantially reduced predicted targetability relative to PAM-only estimates (Figure 3A). In homeostatic microglia, 32.7% (18/55; 95% CI: 20.0–45.5%) of therapeutic genes were predicted targetable by HEAL, compared to 89.1% (49/55; 95% CI: 80.0–96.4%) based on PAM availability alone—a reduction of 56.4 percentage points. Equivalently, 31 of 49 HEAL PAM-bearing therapeutic promoters were not predicted targetable after chromatin filtering in the homeostatic state. SminiCRa showed identical results, as both systems share the Un1Cas12f1 PAM (TTTR). In inflammatory states (naïve, acute LPS, tolerized), HEAL targetability increased to 45.5–49.1% (95% CIs: 32.7–61.8%). The highest values were observed in surgical/stroke-context states from an independent third laboratory: 65.5% (95% CI: 52.7–78.2%) in post-surgical sham and 60.0% (95% CI: 47.3–72.7%) in post-stroke microglia. Because PAM+chromatin targetability is necessarily a subset of PAM-only availability in this framework, these reductions are reported descriptively rather than as a paired hypothesis test (Table S7).

As a genome-wide random-placement sanity check, we also shuffled ATAC-seq peaks across the genome 1,000 times per state while preserving peak size and chromosome distribution. Under random peak placement, only 1.7–3.1% of therapeutic gene promoters overlapped peaks, compared to 65.5–81.8% observed (z-scores 33.0–38.5, all $p < 0.001$; Supplementary Figure S7). This confirms that the observed promoter overlaps are not compatible with uniformly random peak placement, as expected for promoter-enriched ATAC-seq data, but it should not be interpreted as a promoter-matched test of therapeutic-gene-specific enrichment or as validation of CRISPRa activation.

Within each microglial state, predicted targetability differed by 7.3–10.9 percentage points across the six Cas orthologs after chromatin filtering (Figure 3C). By comparison, HEAL targetability varied by 32.7 percentage points across microglial states. Thus, in this promoter-focused atlas, chromatin state accounted for a larger change in predicted targetability than Cas-specific PAM choice. This comparison should be interpreted within the evaluated promoter window and accessibility framework rather than as a general statement about all CRISPRa contexts.

Because the midpoint-based accessibility criterion is intentionally stringent, we also evaluated a relaxed promoter-overlap criterion as a sensitivity analysis (Supplementary Figure S3). Under the relaxed criterion, the fraction of therapeutic promoters overlapping ATAC-seq peaks increased to 65.5–81.8% across states, compared with 32.7–65.5% HEAL PAM+chromatin targetability under the primary criterion. This analysis shows that absolute targetability values depend strongly on how promoter accessibility is operationalized; therefore, the atlas categories should be treated as conservative predictions rather than binary functional outcomes.

2.3 State-dependent chromatin accessibility patterns among therapeutic genes

Analysis of promoter accessibility across the six microglial states revealed four categories of therapeutic genes with distinct implications for CRISPRa design (Figure 4A–B):

Constitutively accessible (21/55 genes): These genes maintained open promoters across all six states and three laboratories, representing the highest-confidence atlas candidates for follow-up. This group includes *Abca1* (cholesterol efflux), *Lamp1* and *Lamp2* (lysosomal membrane), *Mcoln1* (lysosomal calcium channel), *P2ry12* (homeostatic microglial marker), *Tfe3* (autophagy-lysosomal transcription factor), and *Tgfb1* (anti-inflammatory cytokine). These classifications predict promoter accessibility across the surveyed states, but do not establish CRISPRa responsiveness.

Inflammation-gained (8/55 genes): These genes acquired promoter accessibility in inflammatory and/or injury states but had closed promoters in Gosselin homeostatic microglia. This category includes disease-relevant targets such as *Trem2* and *Tyrobp* (phagocytic signaling), *Cx3cr1* (fractalkine receptor), and *Ctsd* and *Ctsb* (lysosomal proteases). These genes represent state-conditional candidates whose predicted targetability depends on the chromatin state of the target microglial population.

Surgical/stroke-context (13/55 genes): These genes showed non-constitutive accessibility patterns that included sham and/or stroke states, while remaining closed in Gosselin homeostatic microglia and variably accessible in LPS-stimulated states. This label is descriptive rather than causal: 12 of 13 genes were accessible in sham, 8 of 13 in stroke, and several were sham-only under the midpoint criterion. This category includes *Il1b* and *Tnf* (pro-inflammatory cytokines), *Grn* (progranulin), *Vegfa* (vascular endothelial growth factor), and notably *Tfeb* (see below).

Never accessible (12/55 genes): These genes had closed promoters in all six states, including *Bdnf*, *Gdnf*, *Il10*, *Pparg*, and *Spi1*. Under the atlas criterion, these targets are not predicted to be promoter-accessible in the surveyed states without additional chromatin remodeling or alternative promoter/TSS selection.

A central case study concerns *Tfeb*, a frequently proposed target for CRISPRa-mediated autophagy activation in microglia [25]. The *Tfeb* promoter met the atlas accessibility criterion in one of six microglial states, post-surgical sham microglia from the Zhang et al. dataset [21] (Figure 4C). This context-dependent classification suggests that *Tfeb* should be treated as a midpoint-sensitive state-context prediction rather than a constitutively accessible CRISPRa target. Post-stroke microglia did not meet the same promoter accessibility criterion, indicating that *Tfeb* targetability may differ between surgical manipulation and ischemic injury models. Because this classification depends on the stringent midpoint criterion, it should not be read as evidence of absolute promoter inaccessibility; available peak files show promoter-overlapping peaks in several non-sham states whose boundaries fall approximately 45–50 bp short of the atlas midpoint. This interpretation remains predictive and requires direct CRISPRa testing.

To explore possible regulators associated with the surgical/stroke-context category, we performed motif enrichment analysis (HOMER) on the 13 promoters using the 21 constitutively open promoters as background. The T-box transcription factor T-bet (Tbx21) was a biologically plausible hit ($p=1\times10^{-4}$, $q=0.0035$; present in 60% of surgical/stroke-context promoters, 0% of constitutive), consistent with IFN-γ/innate immunity signaling as a candidate contributor to post-surgical chromatin remodeling. PRDM1 (Blimp-1), a regulator of terminal immune cell differentiation, was also enriched ($p=1\times10^{-3}$, 50% vs 7%; Supplementary Figure S4). We treat this as exploratory because the analysis was limited to 10 mappable promoters (of 13) and does not establish regulatory causality. Gene Ontology enrichment analysis showed that surgical/stroke-context genes are enriched in

MAPK cascade and STAT tyrosine phosphorylation pathways, while inflammation-gained genes are enriched in microglial cell activation and synapse pruning (Supplementary Figure S5).

ATAC-seq browser tracks at the *Tfeb* and *Tfe3* loci provide locus-level visualization of the atlas classifications (Figure 6). *Tfeb* is called accessible by the midpoint criterion only in post-surgical sham microglia, whereas *Tfe3* shows promoter accessibility across all six states from all three laboratories.

Cross-dataset comparison between Gosselin homeostatic and Zhang sham microglia—both relatively basal but generated by independent laboratories with different protocols—showed 61.8% gene-level concordance (34/55 therapeutic genes). Of the 21 discordant genes, 20 were closed in Gosselin but open in Zhang sham. This pattern is compatible with greater microglial activation in the sham condition, but it may also reflect protocol, depth, sorting, or batch differences and should not be interpreted as cross-dataset validation (Supplementary Figure S6).

In contrast, *Tfe3*—a member of the same MiTF/TFE transcription factor family that regulates an overlapping set of CLEAR (Coordinated Lysosomal Expression and Regulation) network genes [20]—showed constitutive promoter accessibility across all six states and three independent laboratories. For programs seeking to activate the autophagy-lysosomal pathway via CRISPRa in microglia, *Tfe3* is therefore the more consistently accessible atlas candidate under this promoter-accessibility criterion. This should not be interpreted as evidence that TFE3 is functionally superior to TFEB: TFEB and TFE3 are related lysosomal transcription factors whose activity is also shaped by RagA–Folliculin–mTORC1 and post-translational control, so both candidates require direct CRISPRa validation [20, 26].

2.4 A practical framework for state-aware CRISPRa vector design

Based on the atlas, we propose a decision framework for designing CRISPRa vectors targeting microglia (Figure 5A):

1. **Select Cas with packaging constraints in mind.** In the promoter window evaluated here, PAM availability is high ($\geq$89.1% for all Cas in the therapeutic panel, and 100% for SpCas9, CjCas9, and Nme2Cas9), so AAV cargo capacity and activation-domain architecture may be more important practical constraints than PAM coverage alone.

2. **Verify promoter accessibility** in the target cell state using ATAC-seq data (this atlas or equivalent). A gene with abundant PAMs but closed chromatin is not predicted targetable under this atlas criterion.

3. **Design sgRNAs within accessible regions.** Prioritize PAM sites that fall within ATAC-seq peaks, not merely within the promoter window.

4. **Consider state-dependence.** If the target gene is in the "inflammation-gained" category, CRISPRa may only be effective in the disease context—which may be desirable (self-limiting activation) or problematic (ineffective in prevention paradigms).

As a case study, we applied this framework to *Tfe3* (Figure 5B). Using Un1Cas12f1-compatible TTTR PAM sites, we identified candidate protospacers whose gene/Cas combinations are predicted targetable across all six microglial states in the atlas. These sgRNAs are provided as starting points for experimental validation, with state-level atlas flags and heuristic scores reported explicitly as non-validated predictions (Table S3).

3 Discussion

This study presents a cell-type- and state-specific CRISPRa targetability atlas for murine microglia, extending established chromatin-aware sgRNA design principles [14, 17] into a focused resource for a therapeutically relevant cell type. The central observation is that, within this promoter-targeted computational framework, ATAC-seq-derived chromatin accessibility substantially narrows the set of predicted CRISPRa candidates after PAM scanning.

3.1 PAM availability is high in the studied promoter context

The assumption that compact Cas proteins suffer from limited genomic coverage due to restrictive PAMs has influenced vector design decisions in gene therapy. Our data show that, for the promoter windows and curated therapeutic panel evaluated here, PAM availability alone is high. A 350 bp CRISPRa-optimal window provides substantial sequence diversity for at least one suitable PAM site, even for TTTR (expected frequency $\sim$1/128 bp per strand, yielding $\sim$2.7 expected sites per strand and $\sim$5.5 sites when both strands are considered before filtering). The observation that 89.1% of therapeutic gene promoters contain at least one passing HEAL-compatible protospacer is consistent with this sequence-statistical expectation after applying GC, homopolymer, and restriction-site filters.

This does not imply that PAM restriction is irrelevant in all contexts. For applications requiring precise positioning of the Cas-activator complex (e.g., within a narrow 50 bp window), more restrictive PAMs may indeed limit options. However, for standard CRISPRa applications where the optimal window spans hundreds of base pairs, PAM availability may often be less restrictive than chromatin state.

3.2 Chromatin state as the gatekeeper of CRISPRa efficacy

Our integration of ATAC-seq data suggests that chromatin accessibility removes 26.5–63.3% of theoretically PAM-accessible HEAL targets, depending on the microglial state (63.3% in homeostatic, 44.9–49.0% in inflammatory states, and 26.5–32.7% in surgical/stroke-context states, using the 49 PAM-bearing therapeutic promoters as the denominator). This finding aligns with prior observations that CRISPRa efficacy correlates with chromatin accessibility at the target locus [18], but extends them by providing a systematic, genome-wide quantification across multiple microglial states. This qualitative concordance with external CRISPRa evidence gives independent support to the premise that accessible promoters are more plausible CRISPRa targets, while not constituting locus-level validation of the atlas predictions.

The practical implication is that CRISPRa vector design for microglia should include a chromatin accessibility check before prioritizing sgRNAs. Similar state-aware atlases may be useful in other cell types, but the present data support that conclusion directly only for murine microglial promoter-targeted CRISPRa under the framework used here.

3.3 *Tfeb* as a midpoint-sensitive case study and the *Tfe3* comparator

Tfeb is among the most frequently proposed targets for CRISPRa-mediated autophagy activation in neurodegenerative contexts [19, 25]. Our atlas classifies the *Tfeb* promoter as accessible in one of six microglial states—post-surgical sham—and inaccessible under the same midpoint criterion in post-ischemic stroke. This adds a state-dependent hypothesis to its therapeutic tractability, but it should not be read as a binary functional failure call: several non-sham peak files contain promoter-overlapping peaks whose boundaries do not cover the atlas midpoint. We emphasize several caveats: (i) our accessibility classification is based on the midpoint of the promoter window, and partial accessibility at flanking regions may support CRISPRa activity at reduced efficiency; (ii) the sham surgery model involves craniotomy, which differs from spinal cord laminectomy in the degree and type of microglial activation; and (iii) engineered CRISPRa systems with chromatin-remodeling domains might overcome accessibility barriers in closed states.

By contrast, *Tfe3* offers a useful comparator. As a member of the same MiTF/TFE family, TFE3 activates an overlapping set of CLEAR network genes governing autophagy and lysosomal biogenesis [20]. Its constitutive promoter accessibility across all six microglial states from three independent laboratories makes it a more consistently accessible atlas candidate for programs seeking to enhance autophagy-lysosomal function in microglia. However, TFEB and TFE3 form a related lysosomal transcription-factor pair whose activity is shaped by RagA–Folliculin stress-response circuitry and post-translational regulation, not promoter accessibility alone [20, 26]. We therefore present *Tfe3* as a more reliably accessible atlas candidate, not as a functionally superior target. Experimental CRISPRa validation remains required before functional prioritization.

3.4 Limitations

Several limitations should be considered when interpreting this atlas. First, the ATAC-seq datasets analyzed derive from three independent studies using different experimental protocols (ex vivo sorted microglia, in vitro LPS stimulation models), sequencing platforms, depths, FRiP values, and sorting strategies; observed accessibility differences may partially reflect technical batch effects or cell-population differences rather than purely biological variation. For this reason, cross-state rank ordering should be interpreted cautiously; the most robust comparisons are within-dataset contrasts such as sham versus stroke in the Zhang L dataset and cross-dataset consistency of loci such as *Tfe3*. This caveat is particularly relevant for canonical homeostatic markers: *Cx3cr1* and *Tmem119* are classified as closed in the Gosselin homeostatic state and open in later states under the midpoint criterion, whereas *P2ry12* is constitutively accessible. This pattern may reflect limited sensitivity or technical differences in the older homeostatic ATAC-seq dataset rather than true absence of homeostatic regulatory activity; *Tmem119* also lacks a passing TTTR protospacer under the strict Cas12f scan. Second, the promoter accessibility classification based on the midpoint of the CRISPRa-optimal window is a simplification that substantially affects absolute targetability numbers: the stringent HEAL PAM+chromatin criterion yields 32.7% targetability in homeostatic microglia and 32.7–65.5% across all states, while a relaxed criterion requiring only any overlap between a peak and the promoter window yields 65.5–81.8% promoter overlap (Supplementary Figure S3). The relaxed analysis is an accessibility sensitivity analysis, not a guide-level targetability or efficacy estimate. The peak-shuffle analysis is also intentionally simple: peaks were shuffled genome-wide rather than against a promoter-matched or GC-matched null, so it should be interpreted only as a random-placement sanity check, not as evidence that therapeutic promoters are enriched relative to matched promoters or as locus-matched validation of targetability. Third, our sgRNA scoring for Cas12f systems (HEAL, SminiCRa) relies on heuristic filters (GC content, homopolymer avoidance, absence of restriction sites) rather than validated machine-learning models, as no mature equivalent of CRISPick/Rule Set 3 exists for Cas12f [3]. Experimental validation of predicted sgRNA efficacy is required. Fourth, the atlas is based on the murine genome (GRCm39) and microglial ATAC-seq data; translation to human microglia will require integration with human ATAC-seq datasets. Fifth, the observation that PAM availability is high for therapeutic promoters is, in retrospect, predictable from sequence statistics: a 350 bp window searched on both strands provides $\sim$5.5 expected TTTR matches before filtering. Our contribution is the systematic quantification of this expectation across all therapeutic targets and its juxtaposition with state-specific chromatin accessibility. Sixth, this study is purely computational and no CRISPRa predictions have been experimentally validated. While the ATAC-seq accessibility classifications are based on well-established public datasets, confirmation that predicted-open loci are indeed activatable by CRISPRa—and that predicted-closed loci are not—remains essential. Seventh, the 55 therapeutic genes were curated from the microglial biology literature and may be biased toward well-studied genes with more characterized (and potentially more accessible) promoters; genome-wide targetability statistics should be considered alongside the focused panel results. Eighth, alternative TSS usage across microglial states could shift the relevant promoter window for individual genes; our analysis uses a single canonical TSS per gene. Ninth, the microglial sorting strategies differ between datasets: Gosselin et al. used CD11b$^+$ CD45$^{\text{low}}$ gating, which selects resident microglia and excludes infiltrating peripheral macrophages, while Zhang L et al. used CD11b$^+$ CD45$^+$ gating, which may include infiltrating myeloid cells—a known confounder in stroke models where the blood-brain barrier is disrupted. Accessibility differences between Gosselin and Zhang L datasets may therefore partially reflect cell population heterogeneity rather than purely state-dependent chromatin remodeling. Finally, chromatin accessibility is a necessary but not sufficient condition for CRISPRa efficacy—factors such as nucleosome positioning, transcription factor competition, and 3D genome organization also influence activation efficiency.

4 Conclusions

We present a genome-wide CRISPRa targetability atlas for murine microglia that integrates PAM availability for six Cas orthologs with chromatin accessibility across six functional states from three independent laboratories, including a CNS injury model. In this promoter-focused computational framework, PAM availability alone was high across therapeutic genes, whereas ATAC-seq-derived chromatin accessibility substantially narrowed predicted targetability. The atlas classifies *Tfe3* as constitutively accessible across surveyed states, *Tfeb* as midpoint-accessible only in post-surgical sham under the primary criterion, and 13 therapeutic genes as associated with surgical/stroke-context accessibility patterns. This resource is intended as a transparent, hypothesis-generating foundation for state-aware CRISPRa design in microglia, with experimental validation required before therapeutic interpretation.

5 Methods

5.1 Reference genome and gene annotations

All analyses were performed using the GRCm39 (mm39) mouse genome assembly with GENCODE vM33 gene annotations [27]. Protein-coding gene transcription start sites (TSS) were extracted from the GTF annotation file using transcripts tagged as "basic" in GENCODE. For each gene, the most 5' TSS among basic transcripts was selected.

5.2 Promoter window definitions

Two promoter windows were defined for each gene: (1) a wide window of TSS $\pm$ 500 bp (1 kb total), and (2) a CRISPRa-optimal window spanning -400 to -50 bp relative to the TSS. The CRISPRa-optimal window corresponds to the region of maximal CRISPRa efficacy documented in the literature. All coordinates are 0-based, half-open (BED format).

5.3 PAM scanning

Both strands of each promoter region were scanned for PAM sequences of six Cas orthologs: Un1Cas12f1/HEAL (TTTR, 5' PAM, 20 nt spacer), Un1Cas12f1/SminiCRa (TTTR, 5' PAM, 20 nt spacer), SaCas9 (NNGRRT, 3' PAM, 21 nt spacer), SpCas9 (NGG, 3' PAM, 20 nt spacer), Cj-Cas9/MiniCAFE (NNNVRYM, 3' PAM, 22 nt spacer), and Nme2Cas9 (NNNNCC, 3' PAM, 24 nt spacer). IUPAC ambiguity codes were expanded as regular expressions for matching, and reverse-strand searches used IUPAC-aware reverse complements (for example, TTTR reverse-complements to YAAA before reporting the functional PAM orientation as TTTA or TTTG). For each PAM hit, the corresponding protospacer sequence was extracted and filtered for: GC content 30–70%, no homopolymer runs >4 nt, and absence of BsmBI (CGTCTC) and BsaI (GGTCTC) restriction sites and their reverse complements.

5.4 sgRNA scoring for Cas12f

For Cas12f orthologs (HEAL and SminiCRa), no validated machine-learning model for sgRNA efficacy prediction exists. We therefore applied a heuristic scoring system: GC content score (optimal at 40–60%, penalized linearly to 30% and 70%), homopolymer penalty ($\leq$3 nt = 1.0, 4 nt = 0.5, >4 nt = 0.0), and poly-T terminator penalty (TTTTT or longer = 0.0). These components were combined as a weighted average: $S = 0.5 \times S_{\mathrm{GC}} + 0.3 \times S_{\mathrm{homopolymer}} + 0.2 \times S_{\mathrm{polyT}}$, where S_{GC} is 1.0 for GC content 40–60% and decreases linearly to 0.0 at 30% or 70%, $S_{\mathrm{homopolymer}}$ penalizes runs >3 nt, and S_{polyT} is 0.0 if TTTTT or longer is present. Scores are reported as heuristic approximations, not validated efficacy predictions, and experimental validation is required.

5.5 ATAC-seq data sources

Three publicly available ATAC-seq datasets from three independent laboratories were used:

- **Homeostatic microglia:** Gosselin et al. [8]. GEO: GSE89960. Bulk ATAC-seq of ex vivo sorted (CD11b^{+} CD45$^{\mathrm{low}}$) adult mouse microglia. Two biological replicates (SRR5617659, SRR5617660). Single-end 50 bp reads on Illumina HiSeq 4000.

- **Naïve, acute LPS, and tolerized microglia:** Zhang X, Kracht et al. [13]. GEO: GSE175578. ATAC-seq of sorted microglia from mice treated with: PBS+PBS (naïve, PP), PBS+LPS (acute, PL), or LPS+LPS (tolerized, LL), with 1-month interval between treatments. Two technical runs per condition. Paired-end 2×64 bp reads on Illumina HiSeq 2500.

- **Post-surgical sham and post-ischemic stroke microglia:** Zhang, Li et al. [21]. GEO: GSE220041. ATAC-seq of FACS-sorted microglia ($CD11b^{+}$ $CD45^{+}$) from adult mice subjected to sham surgery or transient middle cerebral artery occlusion (tMCAO). Two replicates (sham) and three replicates (stroke), wild-type only (HDAC3-KO samples excluded). Paired-end 2×150 bp reads on Illumina HiSeq X Ten.

5.6 ATAC-seq processing pipeline

ATAC-seq data were processed following established guidelines [6, 28]. Adapter sequences were trimmed using Trim Galore v0.6.11 [29] with Nextera adapter detection. Reads were aligned to GRCm39 using Bowtie2 v2.5.5 [22] with `--very-sensitive` mode; paired-end datasets additionally used `--maxins 1000 --no-mixed --no-discordant`. Aligned reads were sorted and indexed with SAMtools v1.23 [30]. PCR duplicates were removed using Picard MarkDuplicates (v3.0+, REMOVE_DUPLICATES=true); paired-end samples were aligned with read group annotations (–rg-id, –rg SM, –rg LB, –rg PL) to enable proper duplicate detection. Duplication rates were 39.3–40.8% for the Gosselin single-end dataset, 18.2–22.7% for Zhang X paired-end samples, and 12.4–15.1% for Zhang L paired-end samples. Additional filtering removed: reads with MAPQ < 30, mitochondrial reads (chrM), and reads overlapping ENCODE blacklist regions (mm10 blacklist v2 [23], lifted to GRCm39 coordinates using the Python `liftover` package; 18 entries with inverted coordinates after liftover were excluded). Blacklist intersection was performed with BEDTools v2.31 [31]. We note that the standard Tn5 transposase insertion offset correction (+4 bp on the positive strand, −5 bp on the negative strand [6]) was not applied, as the targetability classification relies on peak-level overlap with promoter windows ($\geq$350 bp) rather than base-pair resolution summit analysis; a 4–5 bp shift does not materially affect peak–promoter overlap at this scale.

Technical replicates within each condition were merged using SAMtools. We note that the Zhang X/Kracht dataset (GSE175578) contains two technical runs per condition, not biological replicates; these were merged to increase sequencing depth but do not provide independent biological replication.

Quality control metrics

ATAC-seq quality was assessed using multiple ENCODE-recommended metrics (Supplementary Figure S1). Fraction of Reads in Peaks (FRiP) ranged from 0.27 to 0.57 across samples: homeostatic 0.30, naïve 0.29, acute LPS 0.27, tolerized 0.27, sham 0.57, stroke 0.45. The Zhang X/Kracht samples (PP, PL, LL) fell below ENCODE's recommended threshold of 0.3, consistent with the older sequencing platform (HiSeq 2500, PE 2×64bp); Zhang L and Gosselin samples met or exceeded the threshold. TSS enrichment scores ranged from 9.19 to 13.11 (all >6, ENCODE threshold), consistent with promoter-enriched chromatin accessibility signal. Fragment size distributions for paired-end samples showed the expected nucleosomal banding pattern characteristic of successful ATAC-seq experiments (Supplementary Figure S1C). Irreproducible Discovery Rate (IDR) analysis on biological replicates showed 64.5% (Gosselin), 64.7% (Zhang sham), and 69.4% (Zhang stroke) of peaks passing IDR < 0.05, indicating good inter-replicate reproducibility. IDR was not computed for Zhang X/Kracht samples as these represent technical, not biological, replicates.

Peak calling

Peaks were called using Genrich v0.6.1 [24] in ATAC-seq mode (`-j` flag for ATAC-seq, `-y` to keep unpaired reads for the single-end Gosselin dataset). Genrich was used in place of MACS2 due to a library compatibility issue (undefined symbol `__log_finite`) in the MACS2 Cython extension, which persisted across three Python versions (3.9, 3.10, 3.11) and both conda and pip installations, indicating a system-level `libm/glibc` incompatibility rather than a package-specific bug. Genrich is specifically designed for ATAC-seq peak calling and is used in established pipelines (nf-core/atacseq). We acknowledge the lack of direct Genrich–MACS2 benchmarking on our data as a limitation and recommend that users verify peaks at individual loci of interest using their preferred peak caller.

5.7 Targetability classification

For each gene × Cas × microglial state combination, targetability was determined by two criteria: (1) the midpoint of the CRISPRa-optimal promoter window (−225 bp relative to TSS) must overlap an ATAC-seq peak, establishing promoter accessibility; and (2) at least one PAM site with a passing protospacer must fall within an ATAC-seq peak within the promoter window. A gene was classified as *predicted targetable* if both criteria were satisfied, and *not predicted targetable* otherwise. As a sensitivity analysis, we also evaluated a relaxed accessibility criterion where any overlap between an ATAC-seq peak and the CRISPRa-optimal promoter window (-400 to -50 bp) was sufficient to classify a promoter as accessible, regardless of whether the peak covered the midpoint. Under this relaxed promoter-overlap criterion, 65.5–81.8% of therapeutic promoters overlapped ATAC-seq peaks across states (Supplementary Figure S3). We report results using the stringent midpoint criterion throughout the main text, as it is conservative and better reflects the preferred positioning of CRISPRa complexes near the core promoter.

Genes were further classified by their accessibility dynamics across six states: *constitutively open* if accessible in all six states; *inflammation-gained* if closed in Gosselin homeostatic microglia and accessible across the activated/stroke-associated states; *surgical/stroke-context* if a non-constitutive pattern included sham and/or stroke accessibility without implying injury causality; *never accessible* if closed in all six states; and *other pattern* for non-canonical state combinations.

5.8 Off-target prediction

For the TFE3 sgRNA case study (Figure 5B), candidate protospacers were screened for potential off-target sites by aligning each 20 nt spacer sequence against the GRCm39 genome using Bowtie (v1.3.1, `-v 3 -a`) to enumerate all genomic loci with ≤ 3 mismatches. Candidates with off-target alignments in annotated exonic or promoter regions (± 1 kb of any TSS) were excluded. This heuristic filter is reported only as a preliminary screen and does not replace validated off-target prediction tools (e.g., Cas-OFFinder, CRISPOR) or experimental off-target profiling (e.g., GUIDE-seq), which would be required before therapeutic use.

5.9 Motif enrichment analysis

Motif enrichment for the surgical/stroke-context category was performed using HOMER v4.11 [32] `findMotifsGenome.pl`. The target set was 13 state-context gene promoters (TSS $\pm$ 500 bp), and the background was the 21 constitutively open gene promoters; genome = mm39; motif length = 8,10,12 bp. Known motif enrichment was assessed using HOMER's curated vertebrate motif database. P-values were computed by hypergeometric test; q-values by Benjamini-Hochberg correction. Of 13 surgical/stroke-context promoters, 10 had mappable sequences (3 excluded due to repetitive or ambiguous regions).

5.10 Gene Ontology enrichment

GO Biological Process enrichment was performed using Enrichr [33] (web interface, accessed March 2026) with gene lists from each accessibility category as input. The mouse (Mus musculus) GO Biological Process 2023 library was used. Terms with adjusted $p < 0.05$ (Benjamini-Hochberg) were considered significant.

5.11 Therapeutic gene curation

A panel of 55 genes with documented therapeutic relevance in microglial biology was curated from the literature, spanning seven functional categories: autophagy/lysosome (13 genes), inflammation (10), phagocytosis (6), lipid metabolism (7), neuroprotection (6), microglial identity (9), and epigenetic regulation (4). Each gene was assigned a priority level (critical, high, or medium) based on the strength of evidence for therapeutic modulation in microglia. The full list with justifications is provided in Table S1.

5.12 Computational environment and reproducibility

Analyses were performed using Python 3.11 with a dedicated conda environment containing the required bioinformatics tools. The public GitHub repository listed in the Availability of data and materials section contains the manuscript source, figure-generation scripts, the canonical matrix-level rebuild script, and supplementary tables used for the reported atlas. Rebuilding the atlas from first principles additionally requires external reference/input files not stored in the repository, including the GRCm39 FASTA, GENCODE-derived promoter BED file, ATAC-seq peak files, and raw or processed sequencing files. Raw data accession numbers are GSE89960 (Gosselin), GSE175578 (Zhang X/Kracht), and GSE220041 (Zhang L).

6 Availability of data and materials

The dataset supporting the conclusions of this article is included within the article and its additional files. Public ATAC-seq datasets analyzed in this study are available from the Gene Expression Omnibus (GEO): GSE89960 [8], GSE175578 [13], and GSE220041 [21]. The reference genome (GRCm39) was obtained from UCSC Genome Browser and gene annotations (GENCODE vM33) from the GENCODE database. Analysis scripts, figure-generation code, ranked sgRNA candidate tables, and the complete gene × Cas × state matrix are available at `https://github.com/Jefemaestro33/Paper0-CRISPRa-Targetability-Atlas`. The rendered Figure 6 browser tracks are included in the repository; the underlying bigWig files are large derived intermediates generated from the public ATAC-seq data and are not tracked in GitHub. They can be regenerated from the public GEO inputs using the documented ATAC-seq processing workflow and placed under `analysis_results/bigwigs/` for local browser-track regeneration. An interactive web tool for querying the atlas is under development.

List of abbreviations

AAV: adeno-associated virus; ATAC-seq: Assay for Transposase-Accessible Chromatin using sequencing; CRISPRa: CRISPR-mediated transcriptional activation; CLEAR: Coordinated Lysosomal Expression and Regulation; CNS: central nervous system; DAM: disease-associated microglia; FRiP: Fraction of Reads in Peaks; IDR: Irreproducible Discovery Rate; ITR: inverted terminal repeat; PAM: protospacer-adjacent motif; SCI: spinal cord injury; sgRNA: single-guide RNA; tMCAO: transient middle cerebral artery occlusion; TSS: transcription start site

Declarations

Ethics approval and consent to participate

Not applicable. This study uses only publicly available data and computational analyses.

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

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Authors' contributions

E.D.Z. conceived the study, designed and implemented the computational pipeline, performed all analyses, and wrote the manuscript.

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Additional files

Additional file 1 (CSV; Additional_file_1_Table_S1_therapeutic_genes.csv): Table S1, curated panel of 55 therapeutic genes with functional categories, therapeutic relevance, and priority levels.

Additional file 2 (ZIP-compressed TSV; Additional_file_2_Table_S2_targetability_full.tsv.zip): Table S2, complete targetability matrix for all 21,599 protein-coding genes × 6 Cas orthologs × 6 microglial states, including PAM counts, peak overlap status, and targetability classification.

Additional file 3 (CSV; Additional_file_3_Table_S3_sgrna_recommendations.csv): Table S3, ranked sgRNA candidate table with atlas-level state targetability flags and non-validated heuristic scores. These entries are intended as starting points for experimental validation, not validated efficacy rankings.

Additional file 4 (CSV; Additional_file_4_Table_S4_atac_qc.csv): Table S4, ATAC-seq quality control metrics for all samples, including alignment rates, usable read counts, FRiP, TSS enrichment scores, peak counts, and IDR summary.

Additional file 5 (CSV; Additional_file_5_Table_S5_accessibility_dynamics.csv): Table S5, promoter accessibility classification for all 55 therapeutic genes across six microglial states.

Additional file 6 (BED; Additional_file_6_Table_S6a_blacklist_mm10_original.bed): Table S6a, original ENCODE blacklist regions in mm10 (3,435 entries).

Additional file 7 (BED; Additional_file_7_Table_S6b_blacklist_mm39_lifted.bed): Table S6b, mm39-lifted ENCODE blacklist regions (3,395 entries); 40 entries were lost during liftover (unmapped or inverted coordinates).

Additional file 8 (CSV; Additional_file_8_Table_S7_statistical_tests.csv): Table S7, statistical support table including bootstrap confidence intervals, descriptive PAM-to-chromatin loss counts, and peak-shuffle random-placement results.

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Table 1: Cas orthologs evaluated in the CRISPRa targetability atlas.

Cas	**PAM**	**Spacer (nt)**	**Size (kb)**	**Single-AAV**	**System**
Un1Cas12f1	TTTR	20	∼1.6	Yes	HEAL
Un1Cas12f1	TTTR	20	∼1.6	Yes	SminiCRa
CjCas9	NNNVRYM	22	∼2.9	Yes*	MiniCAFE
SaCas9	NNGRRT	21	∼3.2	No†	—
Nme2Cas9	NNNNCC	24	∼3.2	No†	—
SpCas9	NGG	20	∼4.1	No	—

*At packaging limit with modest activation domains. †Possible with minimal activation domains (e.g., VP64 alone). R = A or G; V = A, C, or G; Y = C or T; M = A or C; N = any. HEAL and SminiCRa use the same Cas protein (Un1Cas12f1) with different engineered sgRNA scaffolds and activation domains; both recognize TTTR (TTTA or TTTG) as functional PAM in mammalian cells.

Figure legends

Figure 1. Overview of the CRISPRa targetability atlas. (A) Schematic of the computational pipeline: promoter definition from GENCODE annotations, PAM scanning across six Cas orthologs,

ATAC-seq reprocessing from public datasets, and integration into a state-aware targetability matrix. (B) Summary statistics of the atlas: 21,599 protein-coding genes, 6 Cas orthologs, 6 microglial states, 55 curated therapeutic genes.

Figure 2. PAM availability across therapeutic promoters. (A) Percentage of 55 therapeutic genes with ≥1 passing PAM site in the CRISPRa-optimal promoter window, by Cas ortholog. All Cas achieve ≥89.1% coverage. Bars with bold outlines indicate single-AAV-compatible systems. (B) Heatmap of PAM counts per therapeutic gene per Cas ortholog. (C) Detail of *Tfeb* and selected genes showing abundant PAM availability.

Figure 3. ATAC-seq filtering reduces predicted CRISPRa targetability. (A) Paired barplot comparing PAM-only targetability (89.1% for HEAL) versus PAM+chromatin targetability (32.7–65.5%) across microglial states. (B) Heatmap of integrated targetability for selected therapeutic genes across states using HEAL. Green = predicted targetable; beige = not predicted targetable. (C) Line plot showing state-level targetability across Cas orthologs after chromatin filtering.

Figure 4. State-dependent chromatin dynamics among therapeutic genes. (A) Classification of 55 therapeutic genes by accessibility pattern across six microglial states: constitutively open (21), inflammation-gained (8), surgical/stroke-context (13), never accessible (12), and other pattern (1; *Megf10*, accessible in homeostatic and stroke only). (B) Gene lists by category. (C) Accessibility of *Tfeb* (midpoint-open only in sham), *Tfe3* (constitutively open in all 6 states), and *Trem2* (inflammation-gained) across all six states. (D) PAM availability for TFEB and TFE3 to illustrate the dissociation between PAM availability and chromatin accessibility.

Figure 5. Practical framework for state-aware CRISPRa vector design. (A) Decision tree: select Cas with packaging constraints in mind → verify chromatin accessibility → design sgRNAs within accessible regions → consider state-dependence. (B) Case study: TFE3 sgRNA candidate prioritization with Un1Cas12f1-compatible TTTR PAM sites.

Figure 6. ATAC-seq signal at the *Tfeb* and *Tfe3* loci across six microglial states. RPKM-normalized ATAC-seq read pileup signal at the *Tfeb* locus (chr17:48,042,955–48,108,344) and *Tfe3* locus (chrX:7,623,799–7,646,441) across all six microglial states from three independent laboratories. Gray shading indicates the gene body; red shading highlights the CRISPRa-optimal promoter window (-400 to -50 bp relative to TSS). *Tfeb* is called accessible by the atlas midpoint criterion only in post-surgical sham microglia, whereas *Tfe3* is called accessible across all six states.

Signal tracks were generated from merged BAMs using deepTools bamCoverage (10 bp bins, RPKM normalization).

Supplementary figure legends

Supplementary Figure S1. ATAC-seq quality control metrics. (A) Bowtie2 alignment rates for all samples. All samples exceed 92% alignment rate. (B) Usable read counts after filtering (MAPQ$\geq$30, chrM removal, blacklist exclusion). (C) Fragment size distributions for paired-end samples showing the expected nucleosomal banding pattern (sub-nucleosomal <150 bp, mono-nucleosomal $\sim$200 bp, di-nucleosomal $\sim$400 bp). Gosselin is single-end (fragment size analysis not applicable). (D) Fraction of Reads in Peaks (FRiP). ENCODE recommends >0.3; dashed line indicates threshold. Zhang L samples (sham, stroke) and Gosselin exceed or meet threshold; Zhang X samples (PP, PL, LL) fall slightly below. (E) TSS enrichment scores. All samples exceed ENCODE's recommended threshold of 6 (dashed line), confirming genuine chromatin accessibility signal. (F) Irreproducible Discovery Rate (IDR) summary for biological replicates: Gosselin (64.5%), Zhang sham (64.7%), Zhang stroke (69.4%) peaks pass IDR < 0.05. IDR was not computed for Zhang X (technical replicates only).

Supplementary Figure S2. Genome-wide CRISPRa targetability. (A) Distribution of passing PAM counts per gene across all 21,599 protein-coding genes, by Cas ortholog. Boxplots show the CRISPRa-optimal window (-400 to -50 bp). (B) Percentage of all 21,599 genes targetable (PAM in ATAC-seq peak) per Cas ortholog per microglial state.

Supplementary Figure S3. Sensitivity of atlas estimates to accessibility criterion. Comparison of the primary stringent atlas criterion and a relaxed promoter-overlap criterion for 55 therapeutic genes. The stringent criterion requires HEAL PAM+chromatin targetability using the promoter midpoint definition. The relaxed comparison counts any ATAC-seq peak overlap with the CRISPRa promoter window and is therefore an accessibility sensitivity analysis, not a validated guide-efficacy estimate. (A) Paired barplot showing that relaxed promoter-overlap values are higher than stringent targetability values across all six states. (B) Per-gene comparison of midpoint targetability states versus midpoint accessibility states. (C) Summary table and interpretation.

Supplementary Figure S4. Motif enrichment in surgical/stroke-context promoters. (A) HOMER known motif analysis comparing 13 surgical/stroke-context promoters against 21 constitutively open promoters as background. Bars show $-\log10$(p-value); red bars indicate q<0.05. Percentages indicate motif presence in target vs background. T-bet (Tbx21) and PRDM1 (Blimp-1) are biologically plausible exploratory hits. (B) Interpretation notes and limitations (n=10 mappable promoters).

Supplementary Figure S5. Gene Ontology enrichment by accessibility category. GO Biological Process enrichment (Enrichr) for each of the four accessibility categories. Constitutive genes are enriched in transcriptional regulation; inflammation-gained in microglial activation and synapse pruning; surgical/stroke-context genes in MAPK cascade and STAT signaling; never accessible in negative regulation of cytokine signaling.

Supplementary Figure S6. Cross-dataset comparison of Gosselin homeostatic vs Zhang sham accessibility. (A) Gene-level concordance: 34/55 (61.8%) therapeutic genes agree. (B) Direction of discordance: 20/21 discordant genes are CLOSED in Gosselin but OPEN in Zhang sham, a pattern compatible with post-surgical activation and/or technical dataset differences. (C) Summary and interpretation. Jaccard index of peak overlap (1kb bins): 0.457.

Supplementary Figure S7. Random-placement peak-shuffle check for promoter accessibility. (A) For each microglial state, ATAC-seq peaks were shuffled randomly across the genome 1,000 times (BEDTools shuffle, preserving peak size and chromosome). Barplot compares observed fraction of therapeutic gene promoters overlapping ATAC-seq peaks (colored bars) versus the expected fraction under random peak placement (gray bars, mean $\pm$ SD). (B) Summary statistics: observed accessibility ranges from 65.5% to 81.8%, while random expectation is 1.7–3.1% (z-scores 33.0–38.5, all $p < 0.001$). This random-placement check shows that observed promoter overlaps are not compatible with uniformly shuffled peaks, but it is not a promoter-matched enrichment test and does not validate CRISPRa activity.

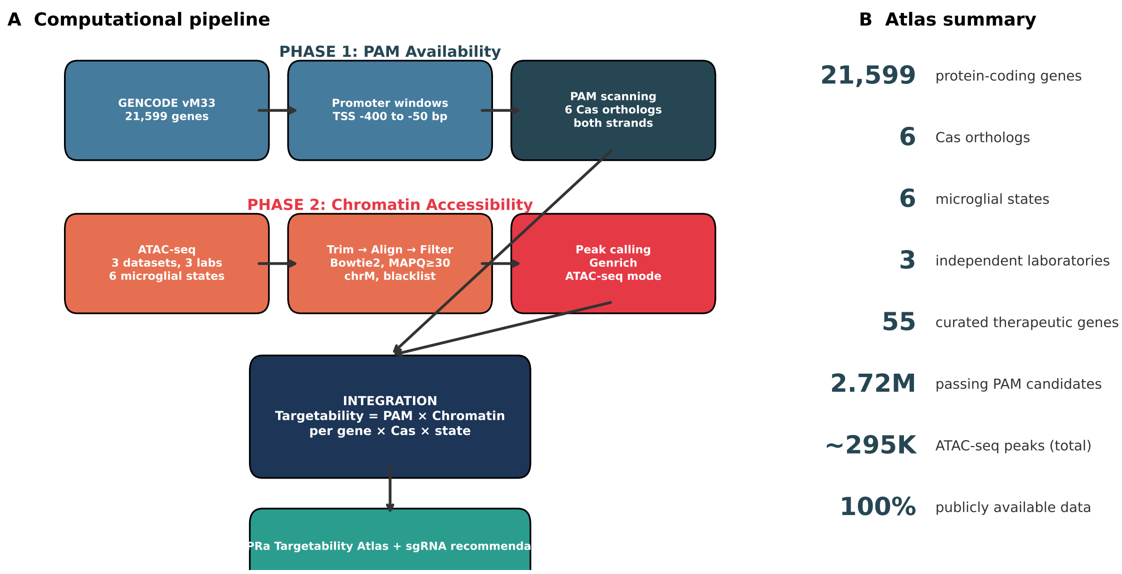


Figure 1

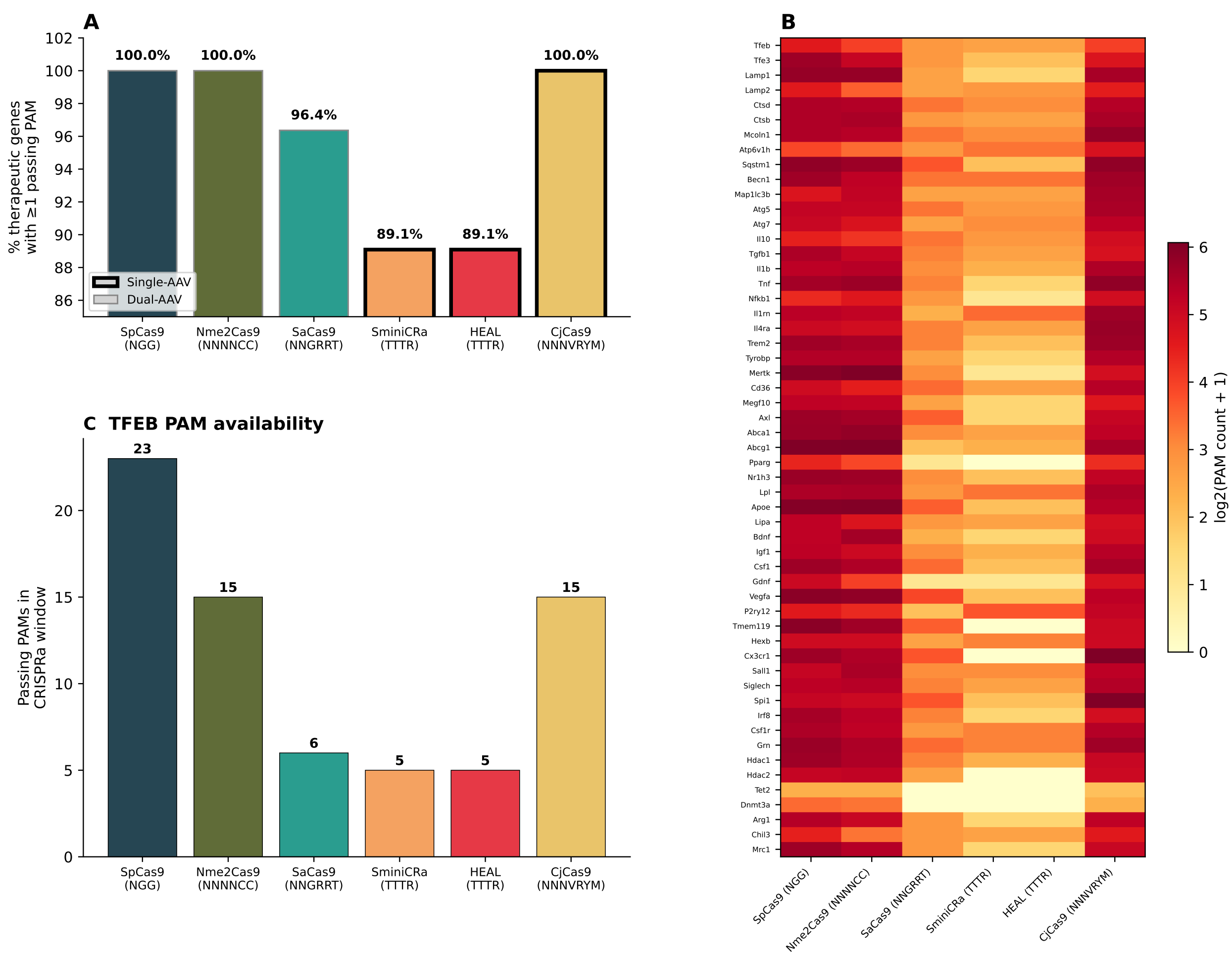


Figure 2

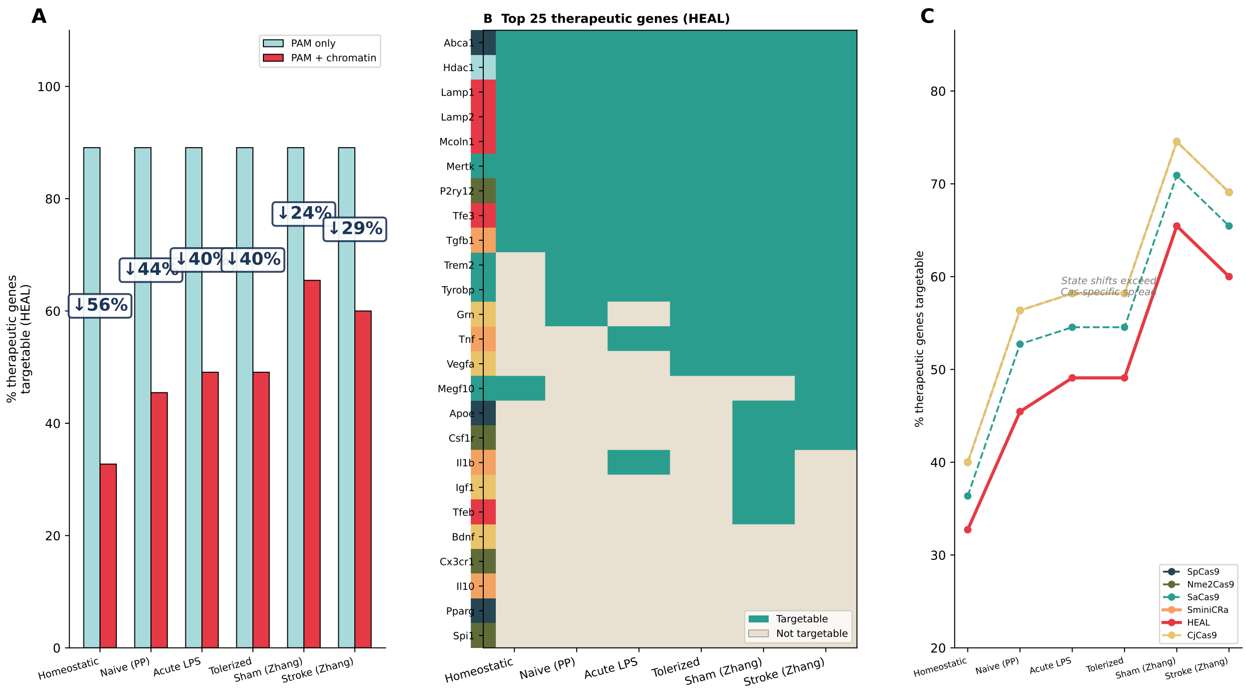


Figure 3

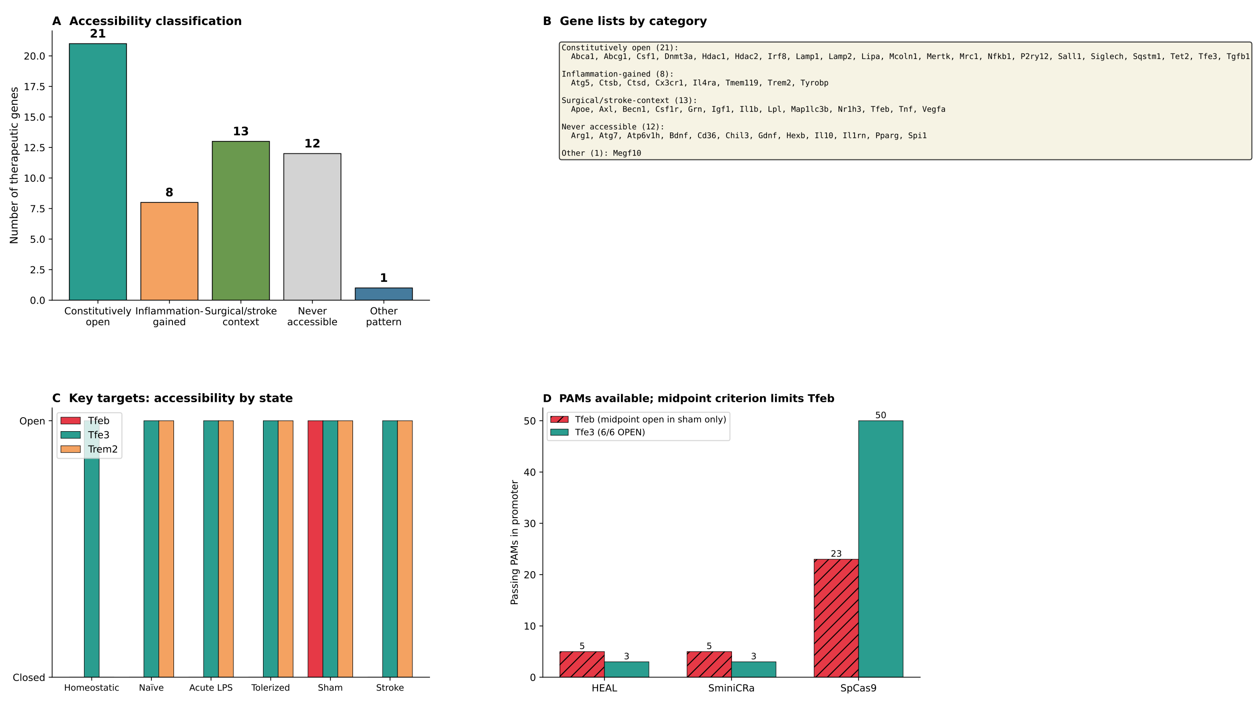
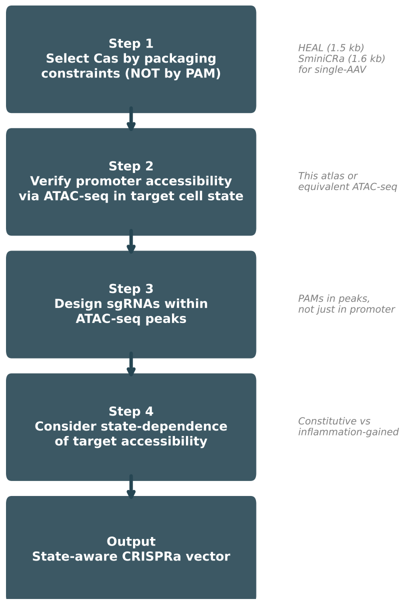


Figure 4

A Decision framework for in vivo CRISPRa design



B TFE3 sgRNA candidates: HEAL (TTTR) system
Constitutively accessible across all microglial states

#	Protospacer (5' to	Strand	GC%	Chromatin
1	GTTGAGCCACGCCTTCATT	−	50%	Open (all states)
2	CGTAGCTTCCTGAGCCACA	−	60%	Open (all states)
3	GCGGAACCATGCCGGCTCT	−	65%	Open (all states)

HEAL system: dUn1Cas12f1 + activation domain in single AAV
All candidates in ATAC-seq peaks across all 6 microglial states (3 labs)
Experimental validation required (heuristic pre-selection, not ML-predicted efficacy)

Figure 5

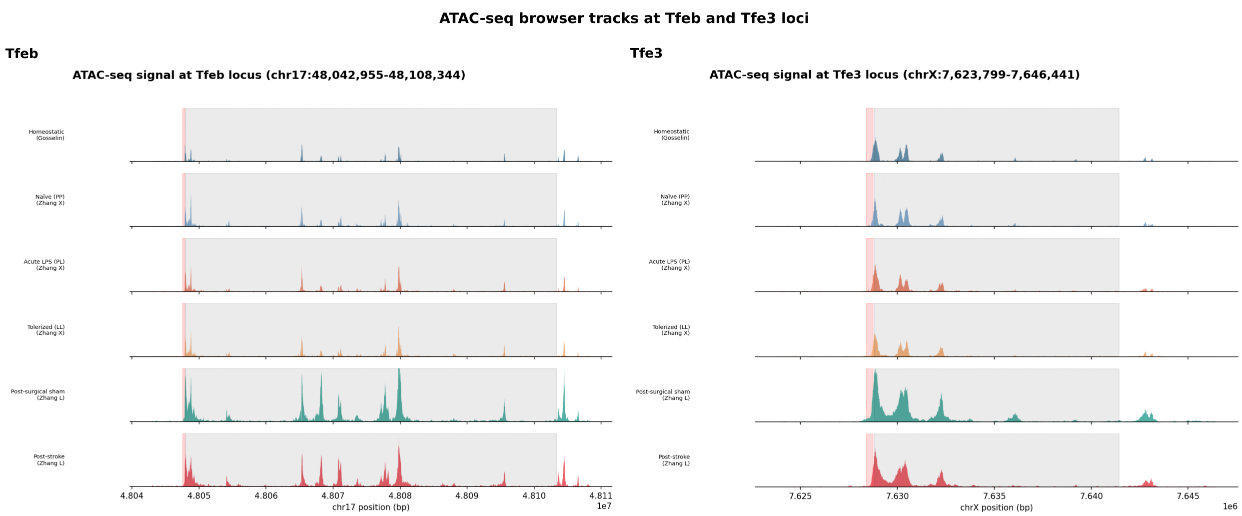
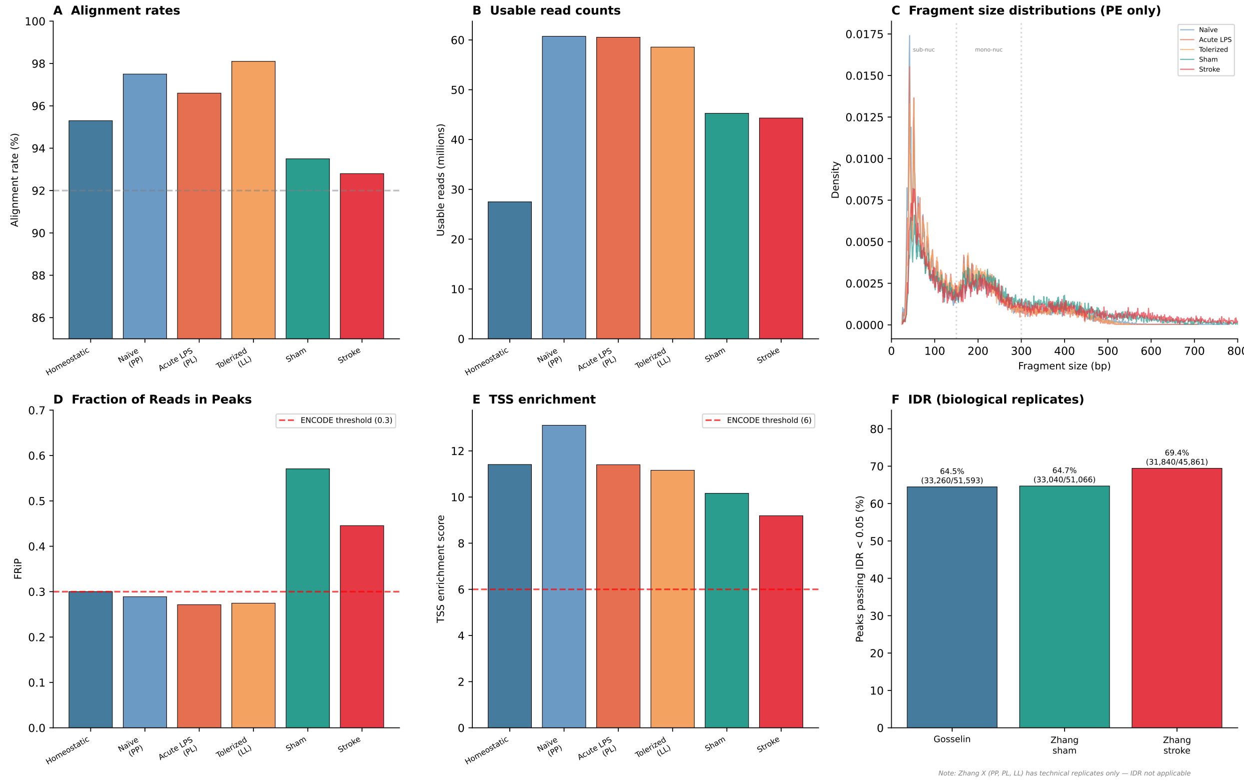
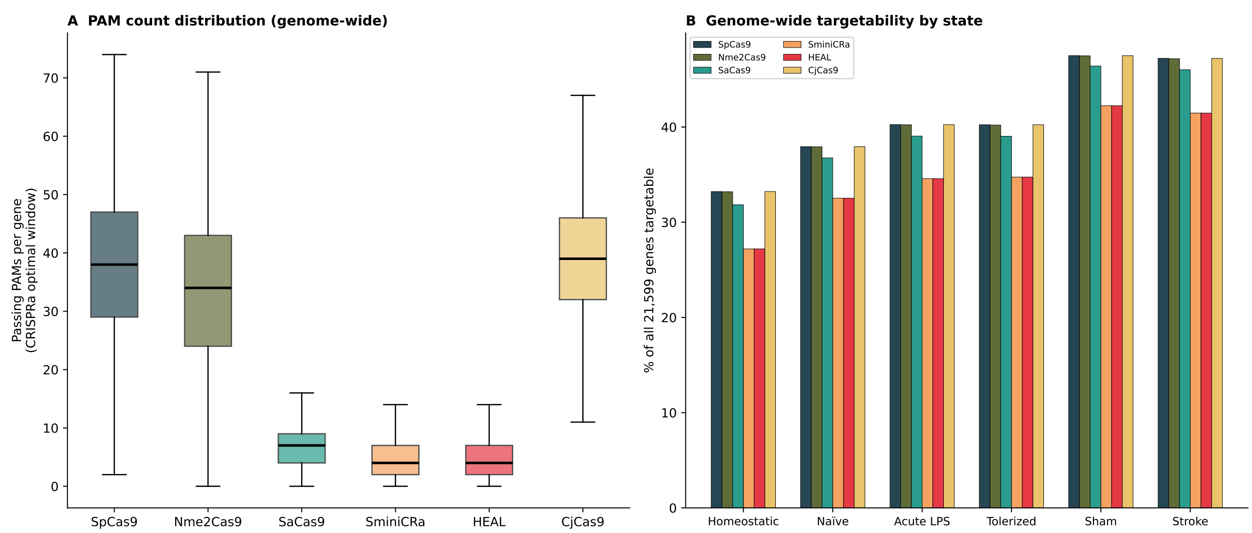


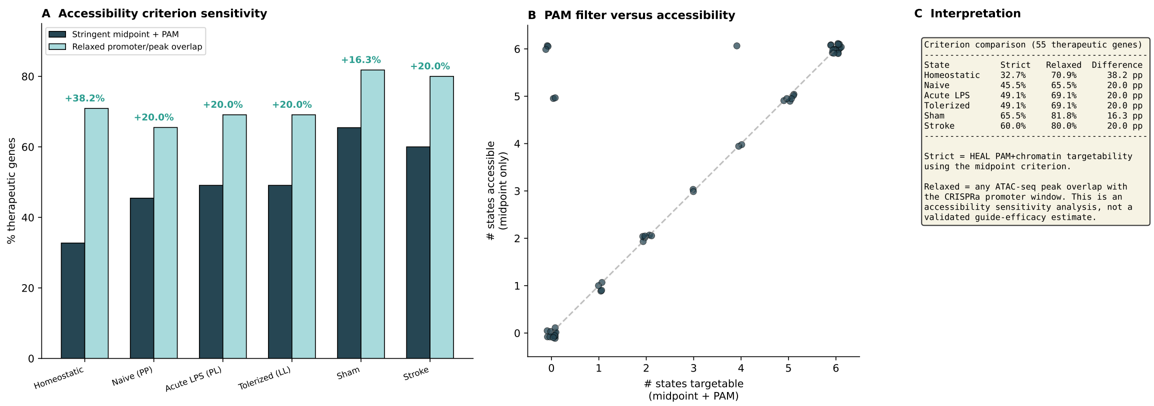
Figure 6



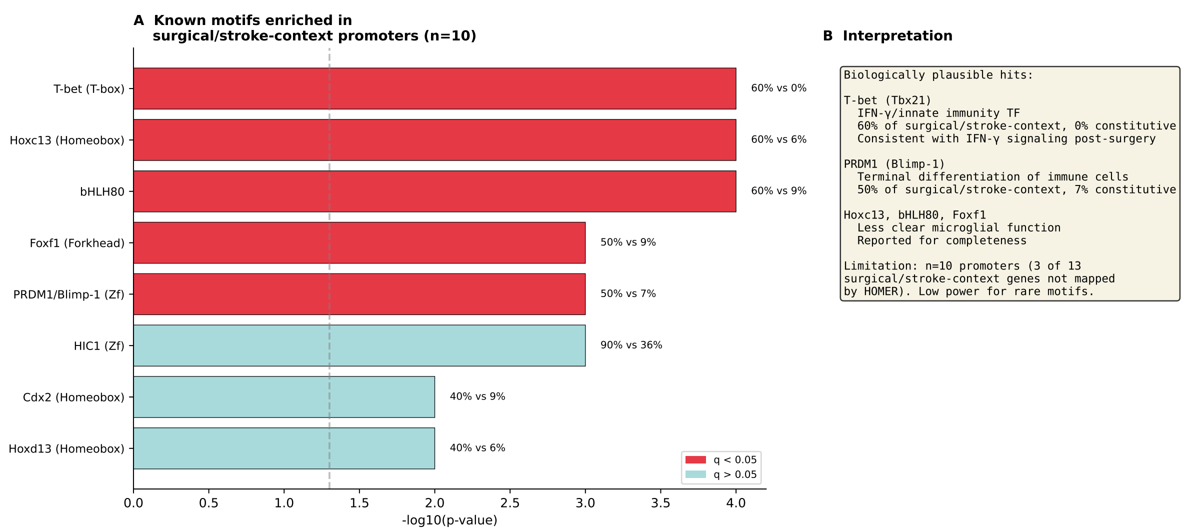
Supplementary Figure S1



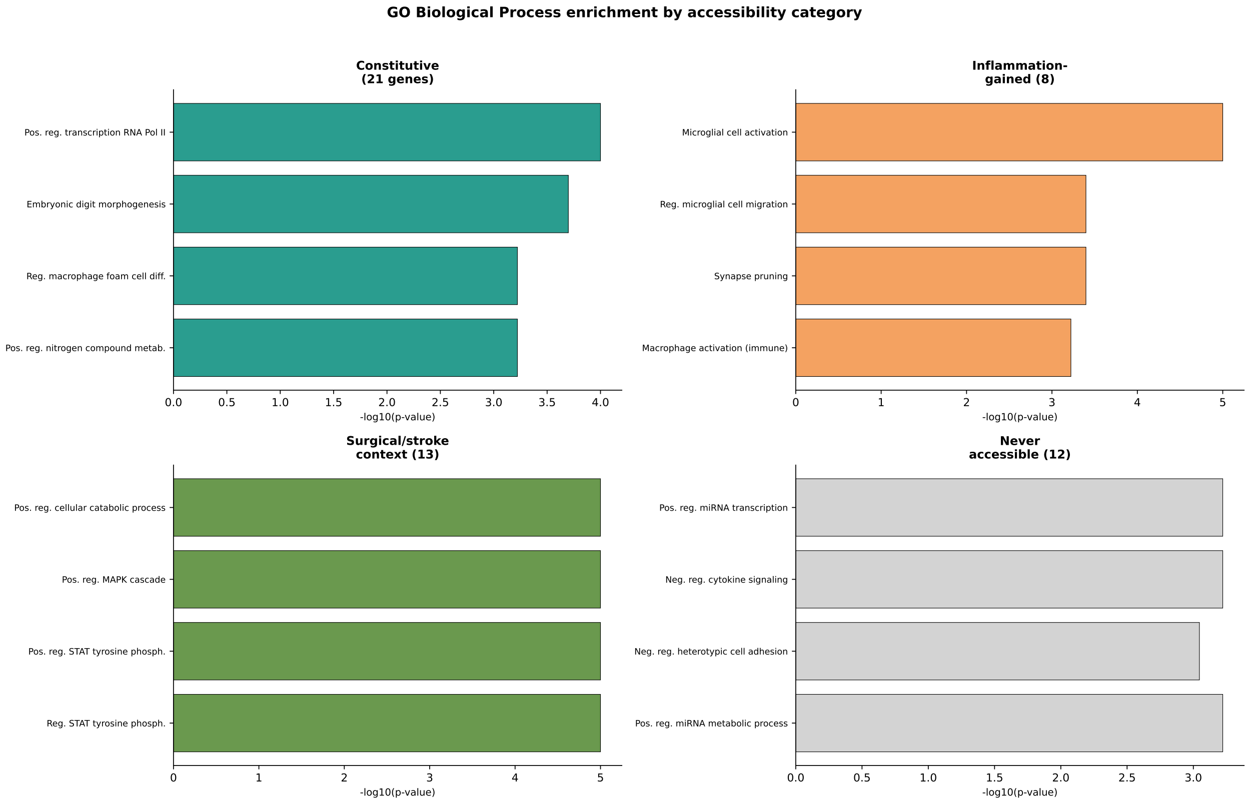
Supplementary Figure S2



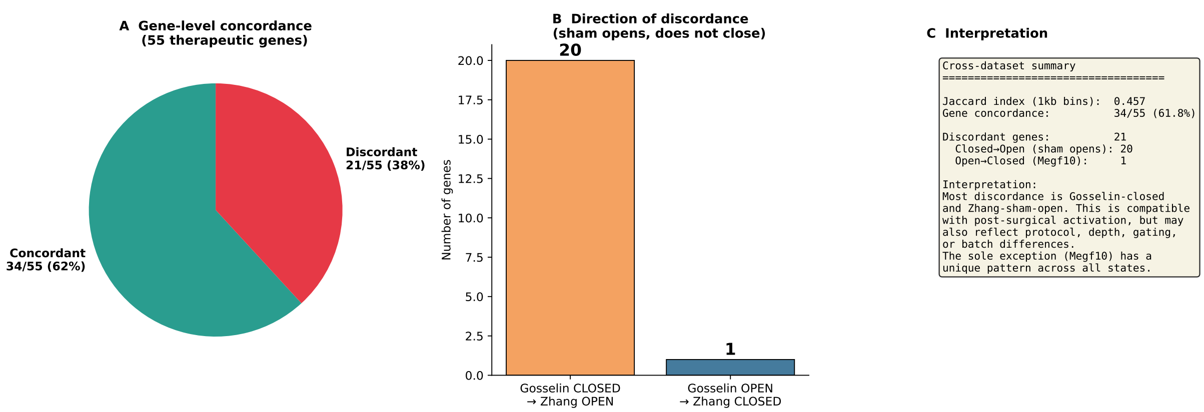
Supplementary Figure S3



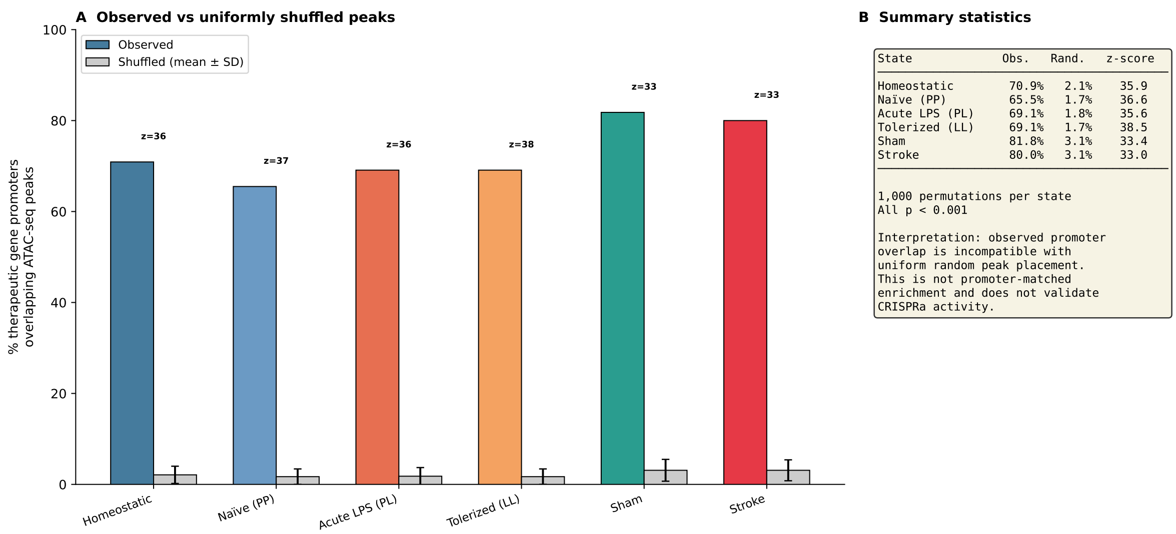
Supplementary Figure S4



Supplementary Figure S5



Supplementary Figure S6



Supplementary Figure S7